

Course outline Summer 2025

Course Name: Data Structures and Algorithms

Course Code: CS-2920

INSTRUCTOR

Instructor(s)	Phoebe Edward Nashed Kirelloss
E-mail	Phoebe.edward@uofcanada.edu.eg
Office Location	B.312
Office Hours	Monday 12:30 pm till 1:00 pm Wednesday 12:30 pm till 1:00 pm Kindly confirm your presence during this slot through email prior to the day of office hours.

TEACHING ASSISTANT

Instructor(s)	Omar Aymen Eltouny
E-mail	omar.aymen@uofcanada.edu.eg
Office Location	B.318
Office Hours	Tuesday 12:00 pm till 1:00 pm

Class Time

Lecture	Lab
Monday 9:00 am till 12:00 pm	Monday 1:00 pm till 4:00 pm
Wednesday 9:00 am till 12:00 pm	Wednesday 1:00 pm till 4:00 pm

Class LOCATION

Lecture	Lab
A.513	A.513
A.513	A.513

COURSE DESCRIPTION

This course continues the study of **data structures**, **recursive algorithms**, **searching** and **sorting** techniques, and general strategies for problem solving. It also introduces **complexity analysis** and complexity classes.

RECOMMENDED TEXTBOOKS

- Data Structures and Algorithm Analysis in Java (6th Edition) by Micheal T. Goodrich by Michael T. Goodrich, Roberto Tamassia, Michael H. GoldWasser

Software requirements

- Java OpenJDK 16 or newer version
- JetBrains IntelliJ

We may also use:

- JavaFX 11
- SceneBuilder

Pre-requisites:

Computer Science 2910 and 6 Math credits - Must be completed prior to taking this course.

LEARNING OBJECTIVES

LO #1 Demonstrate an understanding of fundamental data structures.

LO #2 Develop a thorough understanding of various searching, sorting, and manipulating algorithms.

LO #3 Design and Develop custom Data Structures.

LO #4 Match usage scenarios and applications to appropriate data structures.

CLASS SCHEDULE AND GRADING SCHEME

Week #	Date	Lecture	Lab
1	Wed. 11/6	Abstract Data Types (ADT), Data Structures (DS), Types of Data Structures Java Revision, OOP Revision	Further Exercises on related topic of the week.
	Mon. 16/6	<u>Fundamental Data Structures:</u> Arrays, Singly/Doubly LinkedLists, Cloning DS	Further Exercises on related topic of the week.
2	Wed. 18/6	<u>Recursion</u>	Further Exercises on related topic of the week.
	Mon. 23/6	Quiz 1 – in the beginning <u>Stacks & Queues:</u> Stacks, Queues, Doubly-ended Queues	Further Exercises on related topic of the week.
3	Wed. 25/6	<u>Trees:</u> Trees, Binary Trees, Implementing Trees, Tree Traversal	Further Exercises on related topic of the week.
	Wed. 2/7	Midterm Exam during lecture. <u>Trees</u> (Cont'd)	Further Exercises on related topic of the week.
4	Mon. 7/7	<u>Sorting Techniques</u>	Further Exercises on related topic of the week.
	Wed. 9/7	<u>Priority Queues:</u> Priority Queues (PQ), Implementing PQ, Heaps	Further Exercises on related topic of the week.
5	Mon. 14/7	Quiz 2 – in the beginning <u>Search Trees:</u> Binary Search Trees and AVL Trees	Further Exercises on related topic of the week.

	Wed. 16/7	<u>Graphs:</u> Graphs, Graph ADT, Graph Traversal, Shortest Path	Further Exercises on related topic of the week.
6	Mon. 21/7	Quiz 3 – in the beginning <u>Graphs</u> (Cont'd)	Further Exercises on related topic of the week.
	Wed. 23/7	<u>Revision</u>	Further Exercises on related topic of the week.
	28/7	Study Day	
	29/7 – 31/7	Final Exams	

GRADING SCHEME

<i>Assessments</i>	<i>Grade</i>	<i>Date</i>
<i>Assignments</i>	20%	Ass 1: week 2 - Monday 23/6/2025 Ass 2: week 5 - Monday 14/7/2025 Ass 3: week 6 - Monday 21/7/2025
<i>Quizzes</i>	20% Best 2 Quizzes	Quiz 1: week 2 - Monday 23/6/2025 Quiz 2: week 5 - Monday 14/7/2025 Quiz 3: week 6 - Monday 21/7/2025
<i>Midterm</i>	25%	Midterm in Week 3: during lecture on Wednesday 2/7/2025
<i>Final</i>	35%	During the period of examinations from 29/7 till 31/7
<i>Bonus</i>	5%	3% participation in tutorials. 2% extra problems in midterm/final exams. These extra grades will be added on any other assessment entity in which the student's grade is not full mark.

HOMEWORK/ASSIGNMENTS

- You are expected to complete all weekly reading in preparation of class. Occasionally I may also hand out articles or cases not in the text that are required for you to read and prepare for class discussion. In addition, you will have individual/group assignments and term-based reports and assignments throughout the term. **ALL REQUIRED COURSE WORK MUST BE SUBMITTED TO PASS THE COURSE.**
- All reports required during this course shall be original and following academic integrity policy. The reports will be submitted on Turnitin tool, Turnitin check for similarities among other reports and the internet. If similarity beyond reasonable amount is detected penalties extending to filing an academic dishonesty case may be applied
- All students are expected to submit their work on time, late submissions without prior approved excuse will not be accepted.
- Students are required to use report templates if posted by the instructor.

EXAMS

As per the below assessment dates table, generally, unless otherwise stated by the instructor, midterm exam/s (if applicable) will be held during the regularly scheduled lecture time.

ATTENDANCE AND PUNCTUALITY

Students in this course are required to attend all classes and be on time for all classes. There are no excuses for missing class. All absences are counted as time missed. This applies to all lectures, practical tutorials and/or labs. This also applies to

mandatory office hours for probation & Gateway students. Students miss out on opportunities to acquire academic skills such as critical thinking, problem-solving, oral communication, teamwork, and decision-making when they fail to attend class. Being late for class disrupts the flow of the lecture and is disrespectful to the instructors and fellow students. Accordingly, students should note the following:

1. The instructors are not required to give substitute assignments or make up examinations for students who are absent.
2. It is the student's responsibility to make up the class content missed in consultation with his/her instructor(s). This does not mean that the instructor will provide a substitute class for the student.
3. Grades can be affected by missing a class as the instructor may include class participation marks and/or in-class assignments, or pop-up quizzes as part of the assessment plan for the course.
4. If a student is absent for 25% or more of the class time, tutorials, labs, and/or mandatory office hours, the instructor may:
 - a. Advise the student to drop the course if the absences occur before the drop date for course.
 - b. Give the student a grade of F for the course
 - c. Give the student an Incomplete grade for the course if the absences are caused by an approved documented illness or emergency. Students would then be given the opportunity to complete the course work as per the academic regulation.

ACADEMIC INTEGRITY

All students are expected to conduct themselves in all aspects of this course with the highest level of academic integrity. Accordingly, academic dishonesty will not be tolerated. Regulation 20 of the University Calendar defines academic dishonesty. Where there is reasonable evidence to support an allegation of academic dishonesty, Regulation 20 will be strictly enforced. All students are strongly advised to become familiar with this Regulation and UPEI Cairo Academic Integrity policy which can be found at: [!\[\]\(d3fb9f94af8b26d1c844efa9a98805b0_img.jpg\) UPEI Cairo academic integrity policy.pdf](#)

USE OF AI

The use of generative artificial intelligence (AI) and AI-assisted technologies is permitted only **upon receiving a written approval form the instructor**. Students should disclose in their **assignments** the use of AI and AI-assisted technologies in the writing process. Failure to do so will be considered as violation of the academic integrity policy.